

A Note on the Hyperreals

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This note defines the hyperreal numbers axiomatically, as presented in the literature review by Cutland [1]. We start by extending the ordered field of real numbers.

Definition 0.1. The *extended reals* form an ordered field, $(\hat{\mathbb{R}}, \hat{+}, 0, \hat{\cdot}, 1, \hat{\leq})$, with a proper ordered field inclusion of the reals, $(-)^{\bullet}: (\mathbb{R}, +, 0, \cdot, 1, \leq) \hookrightarrow (\hat{\mathbb{R}}, \hat{+}, 0, \hat{\cdot}, 1, \hat{\leq})$, and extension operations for functions, $(\hat{-}): (\mathbb{R}^n \rightarrow \mathbb{R}) \rightarrow (\hat{\mathbb{R}}^n \rightarrow \hat{\mathbb{R}})$, and for subsets, $(\hat{-}): \wp(\mathbb{R}^n) \rightarrow \wp(\hat{\mathbb{R}}^n)$.

We would like that all properties that hold for the real numbers held for the extended reals as well. This is ensured by assuming the following *transfer principle*. An equivalent, and perhaps more elegant, axiomatisation of the hyperreals was given by Keisler [2] and reviewed by Tall [3].

Definition 0.2 (Transfer principle). An ordered field of extended reals, $(\hat{\mathbb{R}}, \hat{+}, 0, \hat{\cdot}, 1, \hat{\leq})$ with real inclusion $(-)^{\bullet}$, satisfies the *transfer principle* if for every first-order formula, ϕ , on ordered fields, real functions $f_1: \mathbb{R}^{K_1} \rightarrow \mathbb{R}, \dots, f_k: \mathbb{R}^{K_k} \rightarrow \mathbb{R}$, subsets $S_1 \subseteq \mathbb{R}^{M_1}, \dots, S_m \subseteq \mathbb{R}^{M_m}$ and real numbers $r_1, \dots, r_n \in \mathbb{R}$, the closed formula $\phi(f_1, \dots, f_k, S_1, \dots, S_m, r_1, \dots, r_n)$ is true in the ordered field of real numbers if and only if the closed formula $\phi(\hat{f}_1, \dots, \hat{f}_k, \hat{S}_1, \dots, \hat{S}_m, r_1^{\bullet}, \dots, r_n^{\bullet})$ is true in the ordered field of extended real numbers.

Definition 0.3. The hyperreals are extended reals that satisfy the transfer principle.

The extended reals contain infinite and infinitesimal numbers.

Definition 0.4. A number, $x \in \hat{\mathbb{R}}$, is

- *finite* if there is a natural number, $n \in \mathbb{N}$, that bounds it, $-n^{\bullet} \hat{<} x \hat{<} n^{\bullet}$;
- *infinite* if all natural numbers, $n \in \mathbb{N}$, are bounded by it, $n^{\bullet} \hat{<} x$ or $x \hat{<} -n^{\bullet}$;
- *infinitesimal* if it is bounded by the inverses of all natural numbers, $-\frac{1}{n^{\bullet}} \hat{<} x \hat{<} \frac{1}{n^{\bullet}}$ for all $n \in \mathbb{N}$, which we write $x \approx 0$;
- *inifinitely close to* a number $y \in \hat{\mathbb{R}}$ if their difference is infinitesimal, $x - y \approx 0$, which we write $x \approx y$.

Theorem 0.5 (Standard part). *For each finite number, $x \in \hat{\mathbb{R}}$, there is a unique real number, $x^{\circ} \in \mathbb{R}$, that is inifinitely close to it, $x \approx x^{\circ}$.*

Proof. Since x is finite, the set $\{r \in \mathbb{R} : r \leq x\}$ has a supremum in the order of real numbers, $\mathbb{R}$. Thus, we define $x^{\circ} := \sup\{r \in \mathbb{R} : r \leq x\}$. $\square$

The result above defines a monotone function, $(-)^{\circ}: \hat{\mathbb{R}} \rightarrow \bar{\mathbb{R}}$, from the extended reals, $\hat{\mathbb{R}}$, to the ordered set $\bar{\mathbb{R}}$ of real numbers extended with a top, ∞ , and a bottom, $-\infty$, elements.

Definition 0.6. For a real number, $r \in \mathbb{R}$, its *infinitesimal neighbourhood* (or its *monad*) is the set of numbers that are infinitely close to it, $\text{IN}(r) := \{x \in \hat{\mathbb{R}} : x \approx r\}$.

The hyperreals can be shown to be unique once we fix a universe for set theory and a meaning for “al sequences of rationals”. One way to construct them is via ultrafilters.

Example 0.7 (Ultrafilter extended reals). Consider a countably infinite set, N , its filter generated by its cofinite subsets, $\text{cof}(N) \subseteq \wp(N)$, and an ultrafilter that extends this filter, $\mathcal{U} \subset \wp(N)$ with $\text{cof}(N) \subseteq \mathcal{U}$. Consider the set of real functions from the countable set N , $\mathbb{R}^N$, and define an equivalence relation on it that relates two functions if they cannot be distinguished by the sets in the ultrafilter,

$$x \sim y \Leftrightarrow \{n \in N : x(n) = y(n)\} \in \mathcal{U}.$$

The *ultrafilter hyperreals* are defined by this quotient, $\hat{\mathbb{R}} := (\mathbb{R}^N) / \sim$. The reals, $r \in \mathbb{R}$, are included as constant functions, $r^\bullet(n) := r$ for all $n \in N$. The extension operations are defined to preserve the quotient:

$$\begin{aligned} \hat{f}([x_1], \dots, [x_k]) &:= [\langle x_1, \dots, x_k \rangle \circ f] && \text{for } f: \mathbb{R}^k \rightarrow \mathbb{R} \\ \hat{S} &:= \{([x_1], \dots, [x_k]) \in \hat{\mathbb{R}}^k : \{n \in N : (x_1(n), \dots, x_k(n)) \in S\} \in \mathcal{U}\} && \text{for } S \subseteq \mathbb{R}^k \end{aligned}$$

References

- [1] Nigel J. Cutland. Nonstandard measure theory and its applications. *Bulletin of the London Mathematical Society*, 15(6):529–589, 1983. URL: <https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/blms/15.6.529>, doi:10.1112/blms/15.6.529.
- [2] H Jerome Keisler. *Foundations of infinitesimal calculus*, volume 20. Prindle, Weber & Schmidt Boston, 1976. doi:10.2307/2273777.
- [3] David Tall. Elementary axioms and pictures for infinitesimal calculus. *Bulletin of the IMA*, 18:43–48, 1982.